

2. Write the differential equation for the voltage $v_C(t)$ in the circuit in figure E.2 for time $t > 0$, then find an expression for the current $i(t)$ for time $t > 0$.

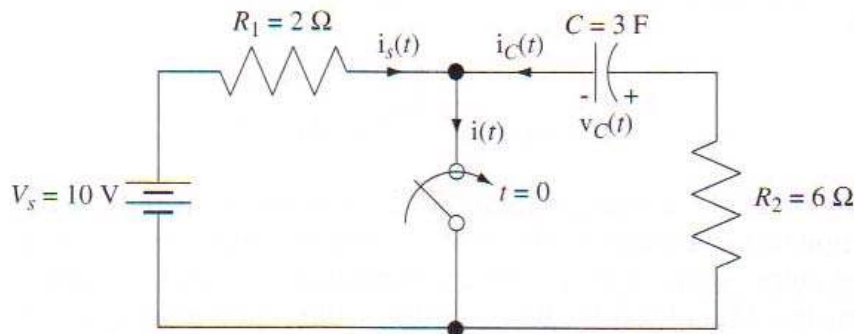


Figure E.2

Answer: $i(t) = 5 + (5/3)e^{-t/18}$

13. Pharmacokinetics is the study of how drugs are absorbed into, distributed through, metabolized by and excreted from the human body. Some drug processes can be approximately modeled by a “one compartment” model of the body in which V is the volume of the compartment, $C(t)$ is the drug concentration in that compartment, k_e is a rate constant for excretion of the drug from the compartment and k_0 is the infusion rate at which the drug enters the compartment.
- Write a differential equation in which the infusion rate is the input signal and the drug concentration is the output signal.
 - Let the parameter values be $k_e = 0.4 \text{ hr}^{-1}$, $V = 20 \text{ l}$ and $k_0 = 200 \text{ mg/hr}$ (where “l” is the symbol for “liter”). If the initial drug concentration is $C(0) = 10 \text{ mg/l}$, graph the drug concentration as a function of time (in hours) for the first 10 hours of infusion. Find the solution as the sum of the zero-input response and the zero-state response.
16. A car rolling on a hill can be modeled as shown in figure E.16. The excitation is the force $f(t)$ for which a positive value represents accelerating the car forward with the motor and a negative value represents slowing the car by braking action. As it rolls, the car experiences drag due to various frictional phenomena that can be approximately modeled by a coefficient k_f , which multiplies the car's velocity to produce a force that tends to slow the car when it moves in either direction. The mass of the car is m and gravity acts on it at all times tending to make it roll down the hill in the absence of other forces. Let the mass m of the car be 1000 kg, let the friction coefficient k_f be $5 \text{ N} \cdot \text{s/m}$ and let the angle θ be $\pi/12$.

- Write a differential equation for this system with the force $f(t)$ as the excitation and the position of the car $y(t)$ as the response.
- If the nose of the car is initially at position $y(0) = 0$ with an initial velocity $[y'(t)]_{t=0} = 10$ m/s and no applied acceleration or braking force, graph the velocity of the car $y'(t)$ for positive time.
- If a constant force $f(t)$ of 200 N is applied to the car, what is its terminal velocity?

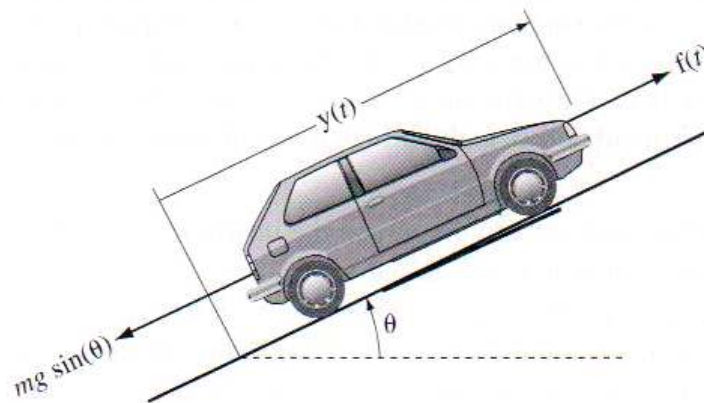


Figure E.16
Car on an inclined plane.

System Properties

- A system is described by the block diagram in figure E.17.

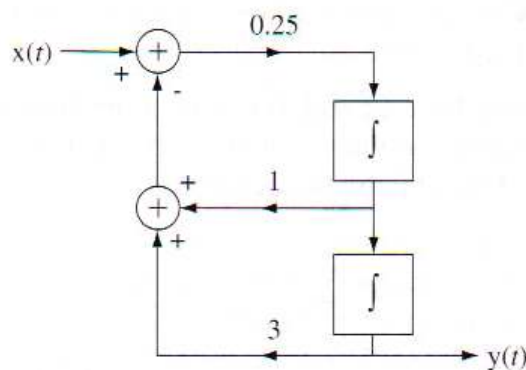


Figure E.17
A system.

Classify the system as to homogeneity, additivity, linearity, time-invariance, stability, causality, memory, and invertibility.

18. A system has an output signal that is the cube of its input signal. Classify the system as to linearity, time-invariance, stability, causality, memory and invertibility.
22. Show that the system described by $y(t) = \text{Re}(x(t))$ is additive but not homogeneous. (Remember, if the excitation is multiplied by any *complex* constant and the system is homogeneous, the response must be multiplied by that same constant.)